

Epidemiology of Sudden Cardiac Death: Clinical and Research Implications

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The current annual incidence of sudden cardiac death in the US is likely to be in the range of 180–250,000 per year. Coinciding with the decreased mortality from coronary artery disease, there is evidence pointing toward a significant decrease in rates of sudden cardiac death in the US during the second half of the twentieth century. However the alarming rise in prevalence of obesity and diabetes in the first decade of the new millennium both in the US and worldwide, would indicate that this favorable trend is unlikely to persist. We are likely to witness a resurgence of coronary artery disease and heart failure, as a result of which sudden cardiac death will have to be confronted as a shared and indiscriminate, worldwide public health problem. There is also increasing recognition of the fact that discovery of meaningful and relevant risk stratification and prevention methodologies will require careful prospective community-wide analyses, with access to large archives of DNA, serum and tissue that link with well-phenotyped databases. The purpose of this review is to summarize current knowledge of sudden cardiac death epidemiology. We will discuss the significance and strengths of community-wide evaluations of sudden cardiac death, summarize recent observations from such studies, and finally highlight specific potential predictors that warrant further evaluation as determinants of sudden cardiac death in the general population.